

Mengdi Wang

Postdoctoral Fellow, University of British Columbia

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Research Summary

My research develops multiscale representations, interface and transport algorithms, and adaptive GPU solvers for physical simulation, while exploring simulation as a two-way interface between AI and the real world: AI expands what can be simulated, and physics provides testable principles for reliable, human-centered intelligent systems.

Research Areas: Physical Simulation, Computer Graphics, Scientific Computing, GPU Computing, Scientific Machine Learning

Academic Appointments

University of British Columbia, Postdoctoral Fellow

2026.9 - Present

• Department of Computer Science, PhysAI Lab, working with Prof. Peter Yichen Chen

Education

Georgia Institute of Technology, Ph.D. in Computer Science

2024.1 - 2026.5

• School of Interactive Computing, advised by Prof. Bo Zhu

Dartmouth College, Ph.D. Program in Computer Science (Transferred)

2020.9 - 2023.12

• Guarini School of Graduate and Advanced Studies, advised by Prof. Bo Zhu

Peking University, B.S. in Computer Science, *summa cum laude*

2016.9 - 2020.6

• Center on Frontiers of Computing Studies, advised by Prof. Baoquan Chen

Publications

WorldAgent: Verification-Guided Agentic Physical World Construction

Submitted to ICLR 2027

Caoliwen Wang*, Mengdi Wang*[†], Yige Chen*, Zejia Wu, Bowen Huang, Siyuan Chen, Guanxiong Chen, Lifu Wei, Heng Zhang, Qinghai Zhang, Yin Yang, Guandao Yang, Shiyong Xiong, Peng Wang, Chenfanfu Jiang, Peter Yichen Chen

* Co-first authors. [†] Corresponding author.

WorldContact: A Contact-Centric World Model for Scalable Robot Learning

Submitted to ICRA 2027

Caoliwen Wang*, Mengdi Wang*[†], Heng Zhang, Shixun Huang, Siyuan Chen, Chao Liu, Anpei Chen, Zhendong Wang, Peter Yichen Chen[†], Huamin Wang

* Co-first authors. [†] Corresponding authors.

WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer

SIGGRAPH Asia 2026

Caoliwen Wang*, Minghao Guo*, Siyuan Chen*, Heng Zhang, Mengdi Wang, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, Peter Yichen Chen

* Co-first authors.

Matrix-Free Multigrid with Algebraically Consistent Coarsening on Adaptive Octrees

Submitted to JCP 2026

Mengdi Wang, Yuchen Sun, Bo Zhu

Cirrus: Adaptive Hybrid Particle-Grid Flow Maps on GPU

SIGGRAPH 2025

Mengdi Wang, Fan Feng, Junlin Li, Bo Zhu (Featured in video trailer)

Real-Time Knit Deformation and Rendering

SIGGRAPH 2025

Mengdi Wang*, Tao Huang*, Haoyang Shi*, Yuxing Qiu, Yin Yang, Kui Wu

* Co-first authors.

An Interface Tracking Method with Triangle Edge Cuts

JCP 2025

Mengdi Wang, Matthew Cong, Bo Zhu

Hydrophobic and Hydrophilic Solid-Fluid Interaction

SIGGRAPH Asia 2022

Jinyuan Liu, *Mengdi Wang*, Fan Feng, Annie Tang, Qiqin Le, Bo Zhu

A Moving Eulerian-Lagrangian Particle Method for Thin Film and Foam Simulation

SIGGRAPH 2022

Yitong Deng, *Mengdi Wang*, Xiangxin Kong, Shiyong Xiong, Zangyueyang Xian, Bo Zhu

A Clebsch Method for Free-Surface Vortical Flow Simulation

SIGGRAPH 2022

Shiyong Xiong, Zhecheng Wang, *Mengdi Wang*, Bo Zhu (Featured in video trailer)

Thin-Film Smoothed Particle Hydrodynamics Fluid

SIGGRAPH 2021

Mengdi Wang, Yitong Deng, Xiangxin Kong, Aditya H. Prasad, Shiyong Xiong, Bo Zhu (Featured in video trailer)

Visual Data Analysis and Simulation Prediction for COVID-19

arXiv preprint (2020)

Baoquan Chen, Mingyi Shi, Xingyu Ni, Liangwang Ruan, Hongda Jiang, Heyuan Yao, *Mengdi Wang*, Zhenhua Song, Qiang Zhou, Tong Ge

Research Experience

Research Intern, Lightspeed Studio, Tencent IEG

2024.5 - 2024.8

- Designed a real-time yarn-level knit fabric simulation framework, resulting in a co-first-author SIGGRAPH 2025 paper.

Research Intern, NVIDIA

2022.6 - 2022.9

- Designed a GPU-accelerated geometric multigrid Poisson solver and VOF-based interface tracking method, published in *Journal of Computational Physics*. Mentored by Ken Museth and Eftychios Sifakis.

Teaching Experience

Instructor, Dartmouth College, COSC 294 (Graduate Seminar)

Fall 2021

Additional Instructor, Dartmouth College / Georgia Tech

Multiple terms

Head Teaching Assistant, Dartmouth College, COSC 277 Computer Graphics

Fall 2023

Academic Service

Reviewer: ACM Transactions on Graphics (SIGGRAPH), Computer Graphics Forum (CGF), Journal of Computational Physics

Awards and Honors

ACM-ICPC Asia Qingdao Regional Contest, Gold Medal (Rank 4th)

2016